

TEACH YOURSELF LECTURE

# Lecture 04

## Data and Analysis

Chapter lecture — 9 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

Computer Science XII · Bilingual support: English + اردو

کمپیوٹر سائنس — بارہویں جماعت: خود سیکھیں، خود پرکھیں

09 teach-yourself lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XII

# How to use this lecture

Learn this chapter **on your own**. Read *Learn it*, try the worked example on paper, then answer *Check yourself* before looking at the answers. The green box repeats the idea in Urdu. ★ Golden topics are high-yield in board exams.

یہ لیکچر خود سیکھیں۔ مثال کاغذ پر حل کریں، پھر جواب دیکھنے سے پہلے خود کو پرکھیں۔

## Lectures in this pack

Chapter 4: Data and Analysis. Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

**L01.** 4.1 Concept of Data Analysis

**L03.** 4.2 Creating a SQLite Database in Python

**L05.** 4.3.3 Handling Missing Values (NaN)

**L07.** 4.5 Data Visualization and Types of Graphs

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**L04.** 4.3 Pandas and DataFrames

**L06.** 4.4 Data Organization and Representation

**L08.** 4.6 Descriptive Statistics

### CHAPTER 4

## Data and Analysis

This chapter shows you how to pull data from a SQLite database with Python, clean it in Pandas, draw the right chart, and summarise a dataset with mean, median, mode and standard deviation. These are the same steps a data analyst uses every day.

اس باب میں آپ سیکھیں گے کہ پائتھن سے ایس کیو لائٹ ڈیٹا بیس کیسے جوڑا جاتا ہے، پینڈاز میں ڈیٹا کیسے صاف ہوتا ہے، درست چارٹ کیسے بنتا ہے، اور اوسط، میڈین، موڈ اور معیاری انحراف سے خلاصہ کیسے نکلتا ہے۔

### What you will learn

- Define data analysis and say why Python is used for it.
- Create a SQLite database, table, insert, select, commit and close.
- Create calculated columns and handle NaN with isnull, fillna and dropna.
- Choose and code bar, line, pie, histogram, scatter and box plots.
- Name the five parts of a Python database connection in order.
- Build a Pandas DataFrame from a dictionary, CSV or Excel file.
- Organise data in tables and frequency distributions.
- Compute mean, median, mode, range, variance, standard deviation, min and max by hand, and know that pandas `.std()` uses  $n-1$ .

## 4.1 Concept of Data Analysis

### LEARN IT

**Data analysis** is examining, organising and interpreting data so you can extract useful information. Schools study results, shops study sales, hospitals study patient records — all to make a decision, spot a trend or solve a problem.

Python is popular for this work because it is readable, talks to files and databases, does statistics, and draws graphs. Five lines of Pandas often replace an hour of spreadsheet clicking.

### WORKED EXAMPLE

A Hyderabad college has 400 CS marks in a spreadsheet. Analysis answers: What is the average? How many failed? Is the second shift weaker than the first? The numbers are data; those answers are information.

### اردو میں سمجھیں

ڈیٹا کا تجزیہ یعنی اعداد کو ترتیب دے کر ان سے فیصلے کے قابل معلومات نکالنا۔ پائنتھن اس لیے منتخب کیا جاتا ہے کہ وہ فائلیں، ڈیٹا بیس، حساب اور گراف سب سنبھال لیتا ہے۔

### CHECK YOURSELF

1. Give one school use of data analysis.
2. Name two Python superpowers for analysis.

Answers: 1. Study results to improve teaching (or similar). 2. Any two: read CSV/Excel/databases, organise tables, statistics, charts.

## 4.1.1 – 4.1.2 Data Sources and Database Connection ★ Golden

### LEARN IT

**Data sources** are where information comes from: school records, websites, sensors, databases.

**Connectivity** is the conversation between your Python program and that source so you can read and update data.

Five components of a database connection, in the order they actually work:

| # | Component         | Role                                                                   |
|---|-------------------|------------------------------------------------------------------------|
| 1 | Database          | Organised store (SQLite = one .db file; MySQL = a server for big apps) |
| 2 | Driver            | Translator between Python and that DBMS (sqlite3 is built in)          |
| 3 | Connection object | Opens, manages and closes the pipeline                                 |
| 4 | Cursor object     | Runs SQL and walks through result rows                                 |
| 5 | SQL queries       | CREATE, INSERT, SELECT, UPDATE, DELETE                                 |

Python → Driver → Connection → Cursor → SQL → Database

### EXAM TIP

Write the five names in order and one line each. A common trap is swapping cursor and connection: the connection is the pipe; the cursor is the messenger that travels through it.

### اردو میں سمجھیں

خمسہ اجزاء: ڈیٹا بیس، ڈرائیور، کنکشن، کرسر، ایس کیو ایل۔ کنکشن پائپ ہے، کرسر اس پائپ میں چل کر حکم چلاتا ہے۔

### CHECK YOURSELF

1. Which object executes `SELECT *`?
2. Is SQLite a driver, a database, or both?

Answers: 1. The cursor. 2. SQLite is the database; the `sqlite3` module is the driver.

## 4.2 Creating a SQLite Database in Python

### LEARN IT

SQLite is built into Python. The whole database is one `.db` file next to your script — no password, no port, no server. Nine steps you must be able to write from memory:

```
import sqlite3
conn = sqlite3.connect("school.db") # 1-2 driver + connection
cur = conn.cursor()                 # 3 cursor
cur.execute("CREATE TABLE IF NOT EXISTS students (id INTEGER, name TEXT, marks INTEGER)")
cur.execute("INSERT INTO students VALUES (1, 'Ali', 85)")
conn.commit()                       # 6 SAVE or the row vanishes
cur.execute("SELECT * FROM students")
rows = cur.fetchall()
for r in rows:
    print(r)
conn.close()                        # 9 unlock the file
```

### OUTPUT

```
(1, 'Ali', 85)
```

| Step     | Why it matters                                     |
|----------|----------------------------------------------------|
| connect  | Creates the file if it does not exist              |
| cursor   | Vehicle that carries SQL                           |
| execute  | Runs one SQL command                               |
| commit   | Writes INSERT/UPDATE/DELETE to disk                |
| fetchall | Brings SELECT results into a Python list of tuples |
| close    | Frees memory and unlocks the file                  |

### COMMON MISTAKE

Forgetting `commit()` is the classic bug: the program prints nothing next time because the insert was never saved. Forgetting `close()` can lock `school.db` so the next run fails.

### REMEMBER IT

Strings inside SQL use single quotes: `'Ali'`. Python strings around the SQL use double quotes (or a triple-quoted string).

اردو میں سمجھیں

ایس کیو لائٹ پائٹھن کے ساتھ آتی ہے، پورا ڈیٹا بیس ایک فائل ہے۔ ترتیب: امپورٹ، کنکٹ، کرسر، EXECUTE، COMMIT، SELECT، FETCALL، CLOSE نہ ہو تو اندراج ضائع ہو جاتا ہے۔

#### CHECK YOURSELF

1. What does \* mean in SELECT \*?
2. Which call must follow INSERT?

Answers: 1. All columns. 2. `conn.commit()`.

## 4.3 Pandas and DataFrames ★ Golden

#### LEARN IT

**Pandas** is a third-party library for tables. Install once: `pip install pandas` in the terminal (not inside the .py file). Convention: `import pandas as pd`.

A **DataFrame** is a 2-D table of rows and columns, like Excel or an SQL table. You can filter, sort and calculate whole columns without writing a `for` loop.

```
import pandas as pd
data = {
    "Name": ["Ali", "Sara", "Ahmed"],
    "Math": [85, 90, 78],
    "Science": [80, 88, 70],
}
df = pd.DataFrame(data)
print(df)
```

#### OUTPUT

|   | Name  | Math | Science |
|---|-------|------|---------|
| 0 | Ali   | 85   | 80      |
| 1 | Sara  | 90   | 88      |
| 2 | Ahmed | 78   | 70      |

`read_csv('students.csv')` and `read_excel('students.xlsx', sheet_name='Sheet1')` load files. `df.head()` prints the first five rows — use it on large files so you do not flood the screen.

```
df["Total"] = df["Math"] + df["Science"]
df["Average"] = df["Total"] / 2
print(df)
```

#### OUTPUT

|   | Name  | Math | Science | Total | Average |
|---|-------|------|---------|-------|---------|
| 0 | Ali   | 85   | 80      | 165   | 82.5    |
| 1 | Sara  | 90   | 88      | 178   | 89.0    |
| 2 | Ahmed | 78   | 70      | 148   | 74.0    |

#### WORKED EXAMPLE

Check: Ali  $85+80=165$ ,  $165/2=82.5$ . Sara  $90+88=178$ ,  $89.0$ . Ahmed  $78+70=148$ ,  $74.0$ .

اردو میں سمجھیں

پینڈا ڈیٹا فریم دیتا ہے۔ ڈکشنری، سی ایس وی یا ایکسل سے لوڈ کریں۔ نئے کالم بغیر لوپ کے پوری قطار پر حساب کر کے بنتے ہیں۔ `head()` صرف پہلی پانچ قطاریں دکھاتا ہے۔

#### CHECK YOURSELF

1. What is `pd`?
2. Why not `print(df)` on a million-row file?

Answers: 1. The usual alias for pandas. 2. It would print everything; use `df.head()`.

### 4.3.3 Handling Missing Values (NaN) ★ Golden

#### LEARN IT

Missing cells are **NaN** ("Not a Number"). Statistics crash or lie if you ignore them. Three tools: `isnull()` (True where empty), `fillna(value)` (replace), `dropna()` (delete the row).

```
import pandas as pd
df = pd.DataFrame({
    "Name": ["Ali", "Sara", "Ahmed", "Ayesha"],
    "Marks": [85.0, None, 78.0, 92.0],
    "Grade": ["A", "A", None, "A"],
})
print(df.isnull().sum())
```

#### OUTPUT

```
Name      0
Marks      1
Grade      1
dtype: int64
```

#### WORKED EXAMPLE

Replace Marks with the mean of the known marks, and Grade with a label. Known marks 85, 78, 92 → mean =  $255/3 = 85.0$ .

```
avg = df["Marks"].mean()          # 85.0
df["Marks"] = df["Marks"].fillna(avg)
df["Grade"] = df["Grade"].fillna("Not Assigned")
print(df)
```

#### OUTPUT

|   | Name   | Marks | Grade        |
|---|--------|-------|--------------|
| 0 | Ali    | 85.0  | A            |
| 1 | Sara   | 85.0  | A            |
| 2 | Ahmed  | 78.0  | Not Assigned |
| 3 | Ayesha | 92.0  | A            |

`df = df.dropna()` would instead delete Sara's row and Ahmed's row (any row that still had a NaN).

#### EXAM TIP

State: identify with `isnull / sum`, fill numeric gaps with the mean, fill categories with a label, or drop rows if filling would invent a lie. Never analyse first and clean second.

#### COMMON MISTAKE

Filling a salary gap with the mean of a company that includes the CEO can hide poverty (the same outlier problem as the mean). Know *why* the value is missing before you fill.

خالی خانے NaN ہیں۔ isnull تلاش، fillna بھرنا (اکثر اوسط سے)، dropna قطار ہٹانا۔ پہلے صاف کرو، پھر تجزیہ۔

## CHECK YOURSELF

1. What does `df.isnull().sum()` return?
2. Does `dropna()` delete a column or a row by default?

Answers: 1. Count of NaNs in each column. 2. Any **row** that contains at least one NaN.

## 4.4 Data Organization and Representation

## LEARN IT

Organised data is easier to read. Three everyday forms:

- **Table** — exact values side by side (Ali 85 Math, 80 Science).
- **Frequency distribution** — how many items fall in each range (bin).
- **Graph** — a picture of a trend or comparison (next topics).

| Range of marks | Frequency (students) |
|----------------|----------------------|
| 50–59          | 5                    |
| 60–69          | 7                    |
| 70–79          | 9                    |
| 80–89          | 4                    |
| 90–100         | 2                    |

Most students scored 70–79. That pattern is hard to see in a raw list of 27 marks, and obvious in this table.

Matplotlib (`import matplotlib.pyplot as plt`) turns tables into graphs: `import` → `figure` → `plot` → `labels` → `title` → `show()`.

ٹیبل درست عدد دکھاتی ہے، تعدد کی جدول بتاتی ہے کتنے طلبہ کس رینج میں ہیں، گراف رجحان دکھاتا ہے۔ میٹپلاٹ لب گراف بناتی ہے۔

## CHECK YOURSELF

1. How many students scored 80 or above in the table?
2. Name the library that draws graphs in Python.

Answers: 1.  $4 + 2 = 6$ . 2. Matplotlib (pyplot).

## 4.5 Data Visualization and Types of Graphs ★ Golden

## LEARN IT

Pick the graph that matches the question. Then label axes and give a title — an unlabelled chart scores zero in an exam.

| Chart        | Shows                                               | Watch out                                                |
|--------------|-----------------------------------------------------|----------------------------------------------------------|
| Bar / column | Compare discrete categories (subjects, departments) | Column = vertical; bar often = horizontal                |
| Line         | Change over time (temperature, visitors)            | Needs an ordered x-axis (months, years)                  |
| Pie          | Parts of one whole as percentages                   | Bad with many or almost-equal slices                     |
| Histogram    | Distribution of one numeric variable in bins        | Bars <b>touch</b> (continuous range), unlike a bar chart |
| Scatter      | Relationship of two numeric variables               | Each dot is one pair (x, y)                              |
| Box plot     | Min, Q1, median, Q3, max, and outliers              | Best tool for spotting extreme values                    |

#### 4.5.1 Bar / column

```
import matplotlib.pyplot as plt
subjects = ["English", "Math", "Science", "History"]
marks = [64, 82, 71, 49]
plt.bar(subjects, marks, color="skyblue", edgecolor="black")
plt.title("Marks vs Subjects")
plt.xlabel("Subjects")
plt.ylabel("Marks out of 100")
```

#### 4.5.2 Line

```
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
temps = [12, 15, 20, 25, 30, 35]
plt.plot(months, temps, marker="o")
plt.title("Average Monthly Temperature")
plt.grid(True)
```

#### 4.5.3 Pie

```
subjects = ["Math", "Physics", "CS", "Biology"]
students = [25, 15, 30, 10] # total 80
plt.pie(students, labels=subjects, autopct="%1.1f%%", startangle=90)
```

Check: Math 25/80 = 31.25% → 31.2%. Physics 18.8%. CS 37.5%. Biology 12.5%.

#### 4.5.4 Histogram

```
scores = [45, 50, 55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82, 85, 90, 95]
plt.hist(scores, bins=[40, 50, 60, 70, 80, 90, 100], edgecolor="black")
```

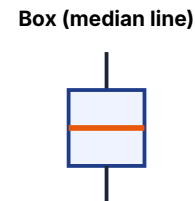
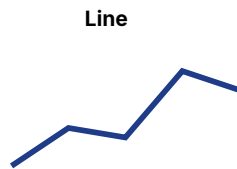
#### 4.5.5 Scatter

```
plt.scatter([10, 20, 30, 40, 50, 60, 70, 80],
            [2, 4, 5, 4, 7, 8, 9, 12], color="green", alpha=0.7)
plt.title("Rainfall vs Crop Yield") # positive correlation
```

## 4.5.6 Box plot

```
plt.boxplot([[55, 60, 65, 70, 75, 80, 85],
             [40, 65, 70, 72, 75, 80, 95]],
            tick_labels=["Class A", "Class B"])
```

The box is the middle 50% (Q1 to Q3). The line inside is the median. Whiskers reach min and max (or a fence, with outliers drawn as dots). Class B is more spread out than Class A.



Column compares; line shows a trend; pie shows share; the box summarises spread around the median.

### EXAM TIP

"Which graph and why?" is a 3-mark gift. Write the name, what it shows, and one reason the others are worse.

اردو میں سمجھیں

بار/کالم اقسام کا موازنہ، لائن وقت کا رجحان، پائی حصے، ہسٹوگرام مسلسل تقسیم (بار آپس میں لگے)، اسکیتھر دو متغیرات کا تعلق، باکس پلاٹ پانچ اعداد اور آؤٹ لائنر۔

### CHECK YOURSELF

1. Why do histogram bars touch?
2. Rainfall vs yield — which chart?

Answers: 1. The x-axis is a continuous numeric range, not separate categories. 2. Scatter (two numeric variables, looking for correlation).

## 4.6 Descriptive Statistics ★ Golden

### LEARN IT

**Descriptive statistics** squeeze a long list into a few honest numbers. **Central tendency** asks "what is typical?". **Spread** asks "how mixed is the list?".

### 4.6.1 Mean, median, mode

| Measure | How                                                                         | Use when                                                  |
|---------|-----------------------------------------------------------------------------|-----------------------------------------------------------|
| Mean    | Sum ÷ n. Uses every value.                                                  | Values sit close together. Example: ordinary test scores. |
| Median  | Middle after sorting. Odd n: centre. Even n: mean of the two centre values. | Outliers would wreck the mean (CEO salary).               |
| Mode    | Most frequent value. May be none, one, or several.                          | Most popular choice (shoe size).                          |

#### WORKED EXAMPLE

Marks 70, 80, 90. Mean =  $240/3 = 80$ .

Odd list 35, 10, 15, 70, 95. Sort: 10, 15, 35, 70, 95. Median = **35**.

Bottles sold 20, 25, 20, 30, 20, 25, 20. Mode = **20** (appears four times).

Outlier trap: nine people earn 10, one earns 1000. Mean =  $1090/10 = 109$  (looks rich). Median = 10 (tells the truth).

#### 4.6.2 Spread: range, variance, standard deviation, min, max

- **Min / max** — smallest and largest. Marks 70, 80, 90, 60, 75 → min 60, max 90.
- **Range** = max – min. Sales 120, 150, 180 → range 60.
- **Population variance**  $\sigma^2$  = (sum of squared deviations from the mean) ÷ n.
- **Sample variance**  $s^2$  uses ÷ (n-1). `pandas.Series.std()` uses the sample version.
- **Standard deviation** =  $\sqrt{\text{variance}}$ . Same units as the data (marks, not marks<sup>2</sup>). Low SD → tight, consistent. High SD → spread out, unpredictable.

#### WORKED EXAMPLE

**Full working** on 70, 80, 90. Mean  $\mu = 80$ . Deviations -10, 0, +10. Squares 100, 0, 100. Sum = 200.

Population variance  $\sigma^2 = 200/3 = 66.67$  (2 d.p.). Population SD  $\sigma = \sqrt{(200/3)} = 8.16$  (2 d.p.).

Sample variance  $s^2 = 200/2 = 100$ . Sample SD  $s = 10$ . That is what `pd.Series([70,80,90]).std()` returns.

#### EXAM TIP

Always state whether you used n or n-1. Show the sorted list for the median. For SD, show: mean → deviations → squares → divide → square root. A box plot pictures min, Q1, median, Q3, max of the same data.

#### COMMON MISTAKE

Do not report variance as if it were in "marks". Variance is marks<sup>2</sup>. Quote the standard deviation when you want a sentence in the original units.

#### اردو میں سمجھیں

اوسط سب کا مجموعہ تقسیم تعداد، میڈین ترتیب کے بعد بیچ، موڈ سب سے زیادہ بار آنے والا۔ رینج بڑا منفی چھوٹا۔ ویرینس اوسط سے فاصلوں کے مربع کی اوسط ہے، معیاری انحراف اس کا جذر ہے اور اصل اکائی میں ہوتا ہے۔ پینڈاز `std()` (نمونے والا  $n-1$ ) استعمال کرتا ہے۔

#### CHECK YOURSELF

1. Median of 4, 8, 6, 10, 2?
2. Range of 120, 150, 180?
3. Does `pandas .std()` divide by n or n-1?

Answers: 1. Sorted 2,4,6,8,10 → 6. 2. 60. 3. n-1 (sample).

## Career Opportunities

### LEARN IT

The skills in this chapter map onto real jobs: data analyst (Pandas, Excel), database administrator (SQLite/MySQL), data scientist (statistics + modelling), visualisation specialist (Matplotlib, Power BI), business-intelligence analyst (dashboards for managers), backend developer (the code that talks to the database).

اردو میں سمجھیں

یہ باب ڈیٹا اینالسٹ، ڈیٹا بیس ایڈمن، ڈیٹا سائنسدان، ویژولائزیشن اور بیک اینڈ ڈویلپر جیسے کیریئرز کی بنیاد ہے۔

## Chapter 4 Review

### Chapter summary

- Data analysis turns raw facts into decisions. Python reads files and databases, calculates and plots.
- Connection chain: database → driver → connection → cursor → SQL.
- SQLite: connect, cursor, execute, commit, fetchall, close. No commit means no save.
- A DataFrame is a 2-D table. Load with a dictionary, `read_csv` or `read_excel`. New columns are vectorised arithmetic.
- NaN: find (`isnull`), fill (`fillna`, often with the mean), or drop (`dropna`).
- Tables give exact values; frequency bins show how many in each range.
- Bar/column = categories; line = time; pie = parts of a whole; histogram = numeric distribution (touching bars); scatter = two variables; box = five-number summary + outliers.
- Mean uses every value and is sensitive to outliers; median is the middle; mode is the most common.
- Range = max–min. Variance = mean squared deviation ( $\div n$  or  $\div n-1$ ). SD =  $\sqrt{\text{variance}}$ . pandas `.std()` uses  $n-1$ .

## Key terms

| Term               | Meaning                                       | اردو                 |
|--------------------|-----------------------------------------------|----------------------|
| Data analysis      | Examining data to extract useful information. | ڈیٹا کا تجزیہ        |
| Connectivity       | Link between a program and a data source.     | رابطہ کاری           |
| Driver             | Translator between Python and a DBMS.         | ڈرائیور              |
| Connection         | Open pipeline to the database.                | کنکشن                |
| Cursor             | Object that executes SQL and fetches rows.    | کرسر                 |
| SQLite             | Built-in file-based database.                 | ایس کیو لائٹ         |
| commit             | Save INSERT/UPDATE/DELETE to disk.            | کمیٹ / محفوظ کرنا    |
| Pandas             | Library for table-like data.                  | پینڈاز               |
| DataFrame          | 2-D labelled table of rows and columns.       | ڈیٹا فریم            |
| NaN                | Missing value ("Not a Number").               | خالی / غیر موجود قدر |
| Frequency          | How many items fall in a class/bin.           | تعدد                 |
| Histogram          | Distribution chart whose bars touch.          | ہسٹوگرام             |
| Scatter plot       | Dots showing two numeric variables.           | اسکیٹر پلاٹ          |
| Box plot           | Min, Q1, median, Q3, max (and outliers).      | باکس پلاٹ            |
| Mean               | Arithmetic average.                           | اوسط                 |
| Median             | Middle value of a sorted list.                | میڈین / درمیانی قدر  |
| Mode               | Most frequent value.                          | موڈ / کثرت           |
| Range              | Max – min.                                    | حد / رینج            |
| Variance           | Mean of squared deviations from the mean.     | ویریانس / تغیر       |
| Standard deviation | Square root of variance (original units).     | معیاری انحراف        |
| Outlier            | An extreme unusual value.                     | آؤٹ لائر             |

## Practice: multiple choice

1. The object that executes SQL is the:  
a. driver  
b. cursor  
c. DataFrame  
d. histogram
2. After INSERT you must:  
a. close then commit  
b. commit  
c. drop the table  
d. call head()
3. Pandas is installed with:  
a. import pandas  
b. pip install pandas  
c. sqlite3.connect  
d. plt.show()
4. `df.head()` shows:  
a. the last 5 rows  
b. the first 5 rows  
c. only NaNs  
d. the standard deviation
5. Missing Pandas values appear as:  
a. NULL in every chart  
b. NaN  
c. 0 always  
d. a pie slice
6. To delete every row that contains a gap:  
a. fillna  
b. dropna  
c. fetchall  
d. commit
7. A line graph is best for:  
a. parts of a whole  
b. a trend over months  
c. two unrelated categories  
d. a primary key
8. Histogram bars:  
a. must have gaps  
b. touch, because bins are a continuous range  
c. always show time  
d. replace the mean
9. Rainfall vs crop yield is a:  
a. pie chart  
b. scatter plot  
c. primary key  
d. mode
10. A box plot's inner line is the:  
a. mean  
b. mode  
c. median  
d. range
11. Mean of 70, 80, 90 is:  
a. 70  
b. 80  
c. 90  
d. 240
12. Median of 10, 15, 35, 70, 95 is:  
a. 10  
b. 35  
c. 70  
d. 45
13. Mode of 20, 25, 20, 30, 20, 25, 20 is:  
a. 25  
b. 30  
c. 20  
d. 7
14. Range of 120, 150, 180 is:  
a. 30  
b. 60  
c. 150  
d. 180
15. pandas `.std()` divides by:  
a. n  
b. n-1  
c. 2n  
d. zero

## Practice: short questions

1. List the five components of a Python database connection in order.
2. Why is `conn.commit()` required after INSERT?
3. What is a DataFrame? How do you load a CSV into one?
4. Differentiate `fillna` and `dropna`.
5. When is a pie chart a poor choice?
6. Differentiate a bar chart and a histogram.
7. Why is the median better than the mean for company salaries?
8. Define range, min and max with the list 70, 80, 90, 60, 75.
9. Why do we take the square root of variance?
10. State two careers that use this chapter's skills.

## Practice: long / problem questions

1. Write a complete Python program that creates `school.db`, a `students` table, inserts Ali 85 and Sara 90, commits, prints all rows, and closes. Number the five connection components beside the matching lines.
2. Starting from the dictionary of Ali/Sara/Ahmed marks, write Pandas code to build a DataFrame, add Total and Average columns, and print it. Show the exact output.
3. Given Marks [85, None, 78, 92] and Grade ["A","A", None, "A"], show `isnull().sum()`, then fill Marks with the mean and Grade with "Not Assigned". Compute the mean by hand.
4. For each goal pick a chart and justify in one sentence: (a) CS vs IT vs Eng enrolment, (b) website hits from Jan to Jun, (c) share of four clubs in one college, (d) distribution of 200 heights, (e) hours studied vs marks, (f) comparing spread of two classes.
5. For 4, 8, 6, 10, 2 compute mean, median, range, population variance and population SD. Show every step. Then say what pandas `.std()` would report instead.
6. Explain, with a numeric example, how an outlier distorts the mean but not the median. Link this to when you would draw a box plot.

## Answer key

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### Multiple choice

1. b
2. b
3. b
4. b
5. b
6. b
7. b
8. b
9. b
10. c
11. b
12. b
13. c
14. b
15. b

## Short questions — model answers

1. Database, driver, connection object, cursor object, SQL queries.
2. Without commit the insert stays in memory and disappears when the program ends, so the next run cannot see the row.
3. A 2-D table of labelled rows and columns. `df = pd.read_csv("file.csv")`.
4. `fillna(v)` replaces NaN with v (the row stays). `dropna()` deletes any row that still contains a NaN.
5. When there are many slices, or the values are almost equal, so the eye cannot compare them.
6. Bar: separate categories, gaps between bars. Histogram: one numeric variable grouped into bins; bars touch.
7. A huge CEO salary pulls the mean up; the median stays with the typical worker.
8. Min 60, max 90, range 30.
9. Variance is in squared units. The SD brings the spread back to the original units (marks, rupees, °C).
10. Any two: data analyst, DBA, data scientist, visualisation specialist, BI analyst, backend developer.

## Long questions — model answers

1. Use the nine-step listing in 4.2 with two INSERTs. Label: `sqlite3` = driver, `connect` = connection, `cursor` = cursor, `execute strings` = SQL, `school.db` = database.
2. Output is the three-row table with Total 165, 178, 148 and Average 82.5, 89.0, 74.0 (see the worked example).
3. `isnull sum`: Marks 1, Grade 1. Mean of 85, 78, 92 =  $255/3 = 85.0$ . After fill: Sara Marks 85.0, Ahmed Grade "Not Assigned".
4. (a) column/bar — compare categories. (b) line — time. (c) pie — parts of a whole. (d) histogram — numeric distribution. (e) scatter — two numeric variables. (f) box — spread and outliers.
5. Sorted 2,4,6,8,10. Sum 30, mean 6, median 6, range 8. Deviations -4,-2,0,2,4. Squares 16,4,0,4,16 sum 40.  $\sigma^2 = 40/5 = 8$ .  $\sigma = \sqrt{8} = 2.828$  (3 d.p.). pandas `.std()` uses  $40/4 = 10$ ,  $\sqrt{10} \approx 3.162$ .
6. Nine salaries of 10 and one of 1000: mean 109, median 10. Draw a box plot — the 1000 appears as an outlier dot above the whisker, so the distortion is visible.

## Self-assessment: I can...

- |                                                                                          |                                                                                            |
|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <input type="checkbox"/> name the five connection components in order.                   | <input type="checkbox"/> write SQLite connect / create / insert / commit / select / close. |
| <input type="checkbox"/> build a DataFrame and add a calculated column.                  | <input type="checkbox"/> find, fill and drop missing values.                               |
| <input type="checkbox"/> choose bar, line, pie, histogram, scatter or box for a problem. | <input type="checkbox"/> compute mean, median, mode, range, variance and SD by hand.       |
| <input type="checkbox"/> state that pandas std uses $n-1$ .                              |                                                                                            |